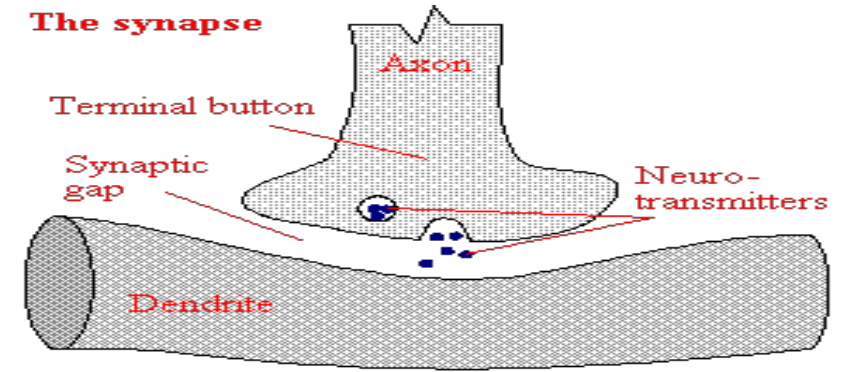
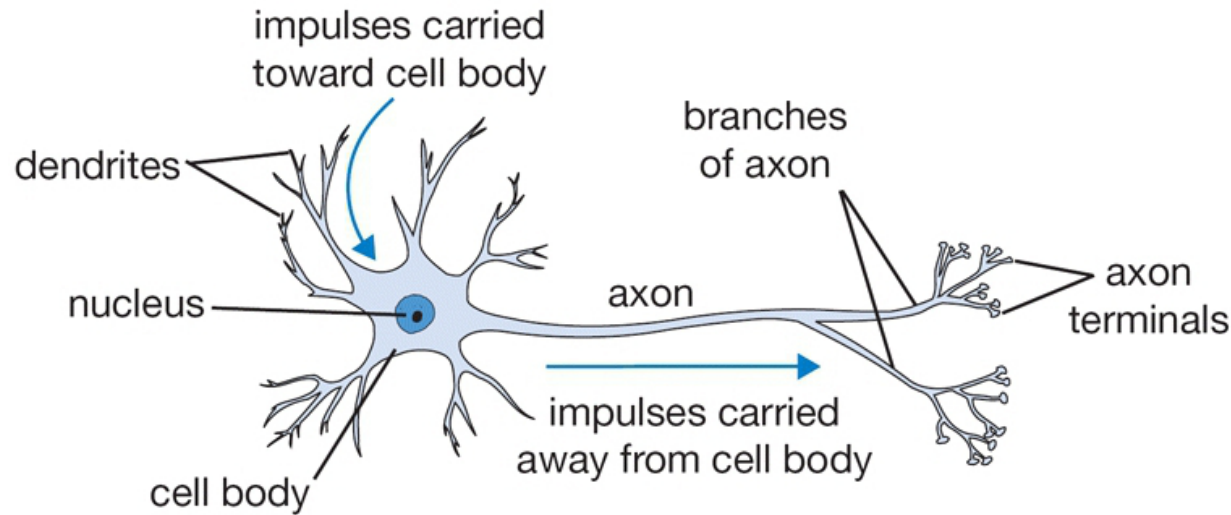


# Artificial Neural Network

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- ❑ An Artificial Neural Network (ANN) is an information processing paradigm that is inspired by the biological nervous systems, such as the human brain's information processing mechanism.
- ❑ An NN is configured for a specific application, such as pattern recognition or data classification, through a learning process. It is composed of a large number of highly interconnected processing elements (neurons) working in unison to **solve** specific problems.
- ❑ Learning in biological systems involves adjustments to the synaptic connections that exist between the neurons. NNs try to model this biological learning process.
- ❑ Most NNs have some sort of **training** rule. In other words, NNs **learn** from examples (as children learn to recognize dogs from examples of dogs) and exhibit some capability for **generalization** beyond the training data.

# Inspired by a biological neuron



- ❑ In the human brain, a typical neuron collects signals from others through a host of fine structures called *dendrites*.
- ❑ The neuron sends out spikes of electrical activity through a long, thin stand known as an *axon*, which splits into thousands of branches.
- ❑ At the end of each branch, a structure called a *synapse* converts the activity from the axon into electrical effects that inhibit or excite activity in the connected neurons.

Image credit: <http://cs231n.github.io/neural-networks-1/>

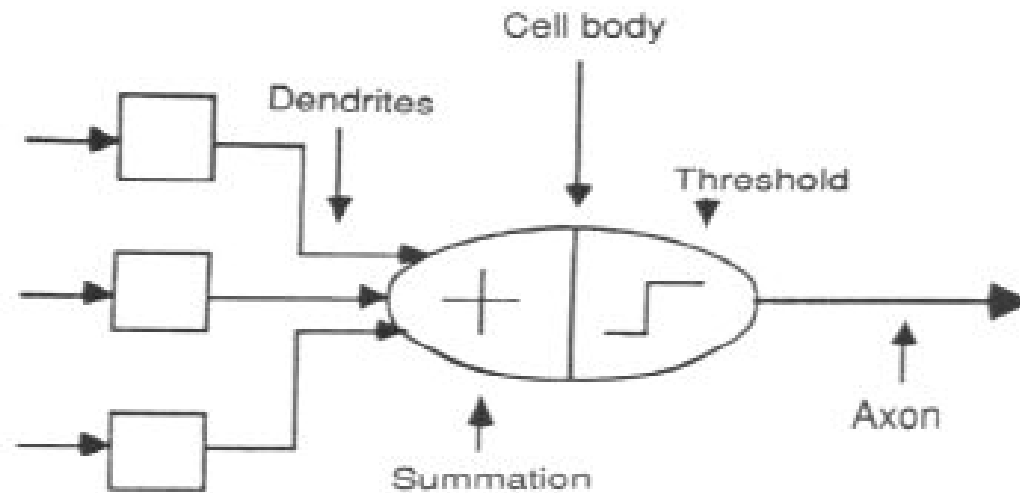
# Biological inspirations

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- ❑ The brain is a collection of about 10 billion interconnected neurons. Each neuron is a cell that uses biochemical reactions to receive, process and transmit information.
- ❑ A neuron's dendritic tree is connected to thousands of neighbouring neurons through synapses. When one of those neurons fire, a positive or negative charge is received by one of the dendrites. The strengths of all the received charges are added together through the processes of spatial and temporal summation.
- ❑ Properties of the brain
  - ❑ It can learn, reorganize from experience
  - ❑ It adapts to the environment
  - ❑ It is robust and fault tolerant

# A Biological Inspired Neuron Model

- ❑ When a neuron receives excitatory input that is sufficiently large compared with its inhibitory input, it sends a spike of electrical activity down its axon.
- ❑ Learning occurs by changing the effectiveness of the synapses so that the influence of one neuron on another changes.



# How to model a single neuron?

